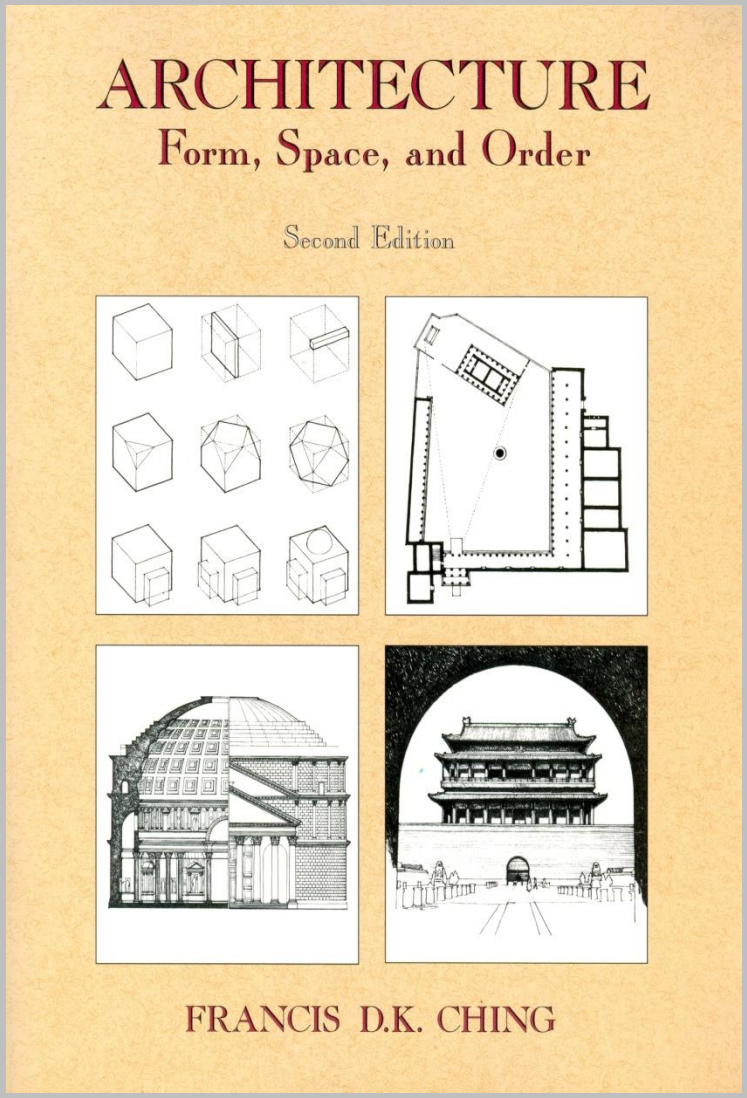
Architecture

Form , Space and Order

ARCH 305: Architecture and Town Planning

PRIMARY ELEMENTS

- Point and Point Elements
- Line and Linear Elements
- Plane and Planer Elements
- Volume and Volumetric Elements



PRIMARY ELEMENTS

"All Pictorial form begins with the point that sets itself in motion … The point moves … and the line comes into being—the first dimension.

If the line shifts to form a plane, we obtain a two-dimensional element.

In the movement from plane to spaces, the clash of planes gives rise to body (three-dimensional) … A summary of the kinetic energies which move the point into a line, the line into a plane, and the plane into a spatial dimension".

PRIMARY ELEMENTS

Point indicates a position

A point extended becomes a

Line with properties of:

- length
- direction
- position

A line extended becomes a

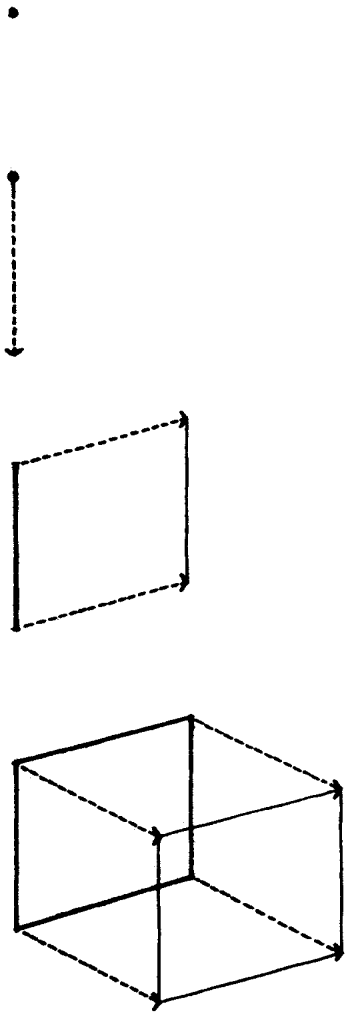
Plane with properties of:

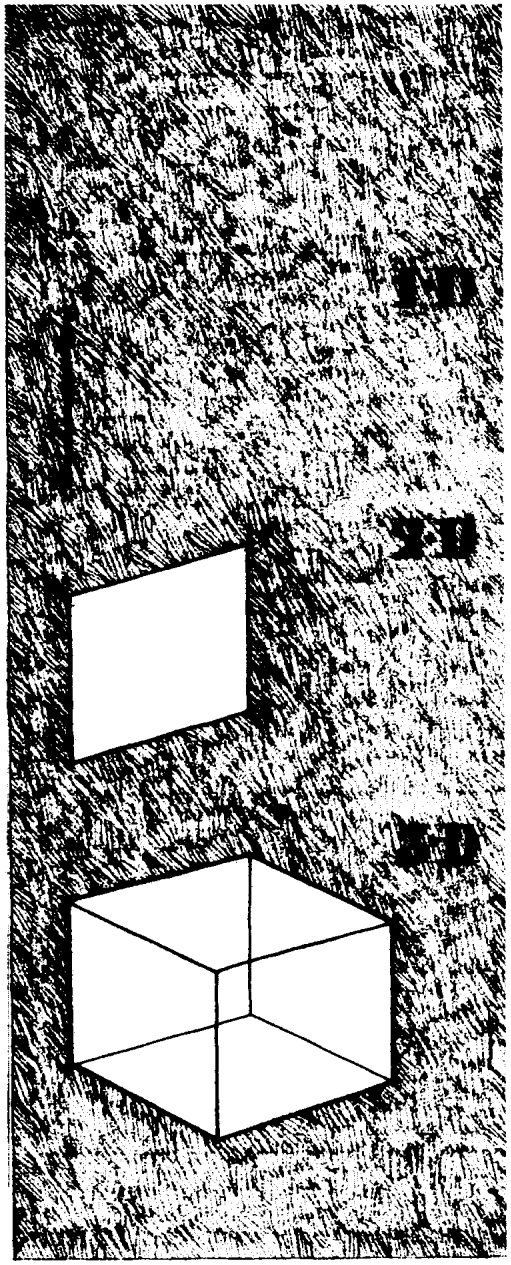
- length and width
- shape
- surface
- orientation
- position

A plane extended becomes a

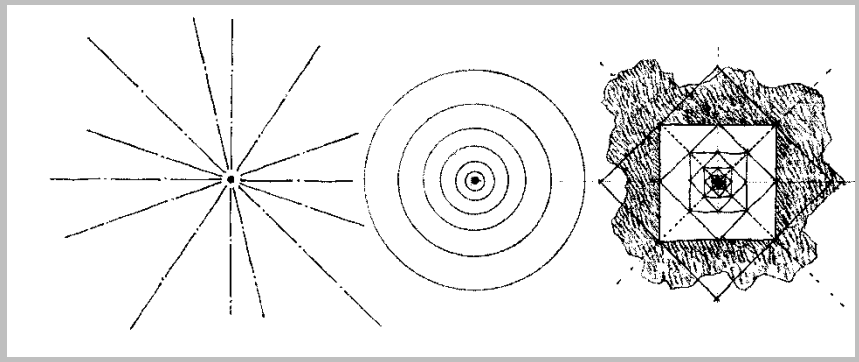
Volume with properties of:

- length, width and depth
- form and space
- surface
- orientation
- position





POINT



A point marks a position in space. Conceptually, it has no length, width or depth, and is therefore static, centralized, directionless.

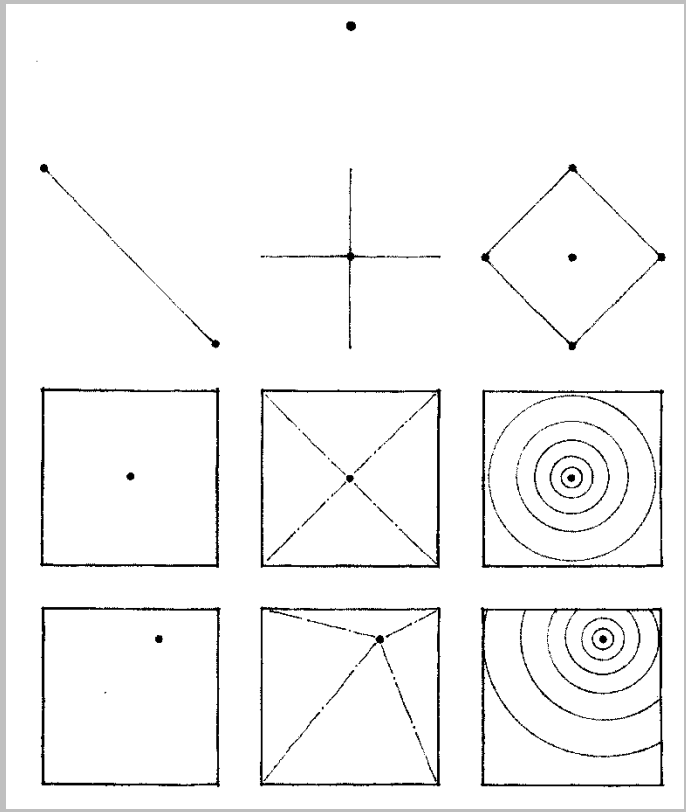
As the prime element in the vocabulary of form, a point can serve to mark:

- the two ends of a line

- the intersection of two lines

- the meeting of lines at the corner of a plane or volume

- the center of a field

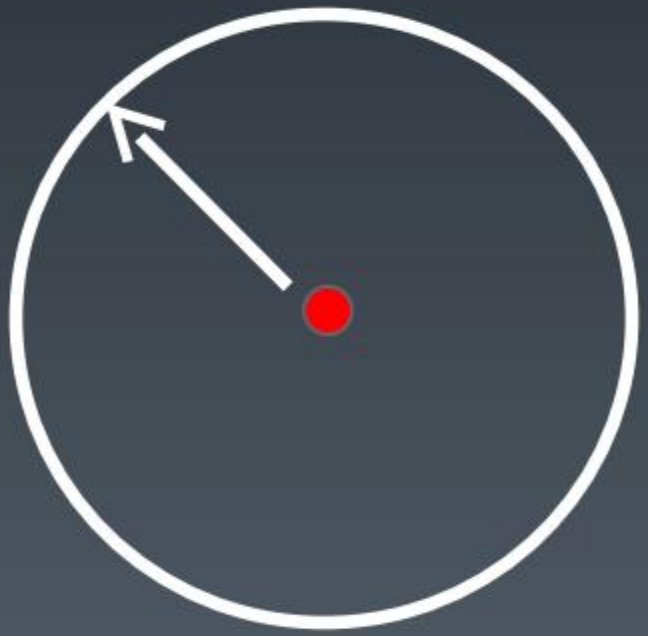


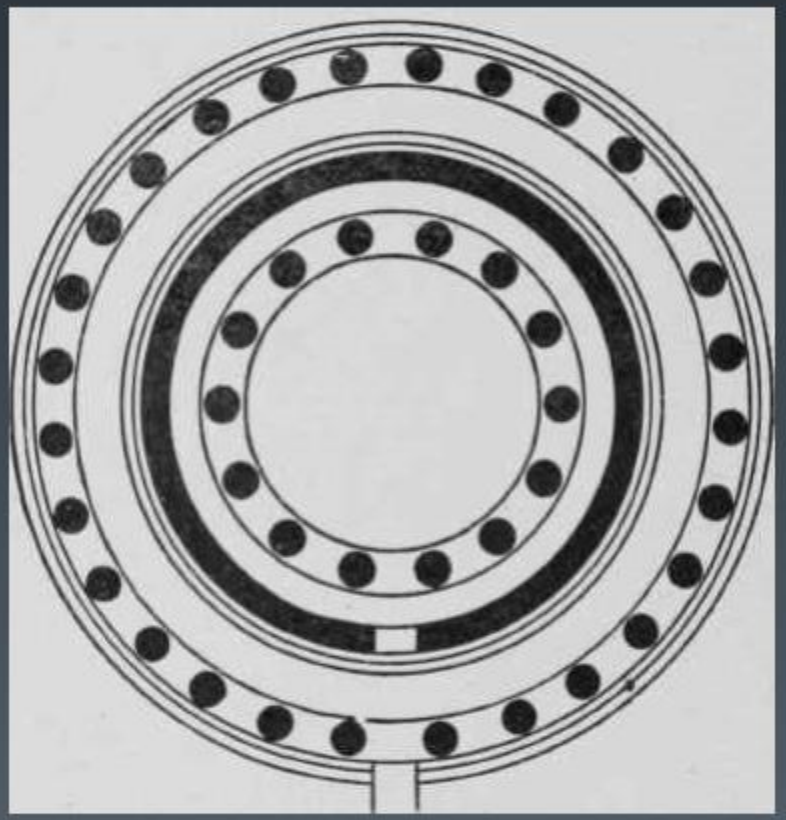
Although a point theoretically has neither shape nor form, it begins to make its presence felt when placed within a visual field. At the center of its environment, a point is stable and at rest, organizing surrounding elements about itself and dominating its field.

When the point is moved off-center, however, its field becomes more aggressive and begins to compete for visual supremacy.

Visual tension is created between the point and its field.

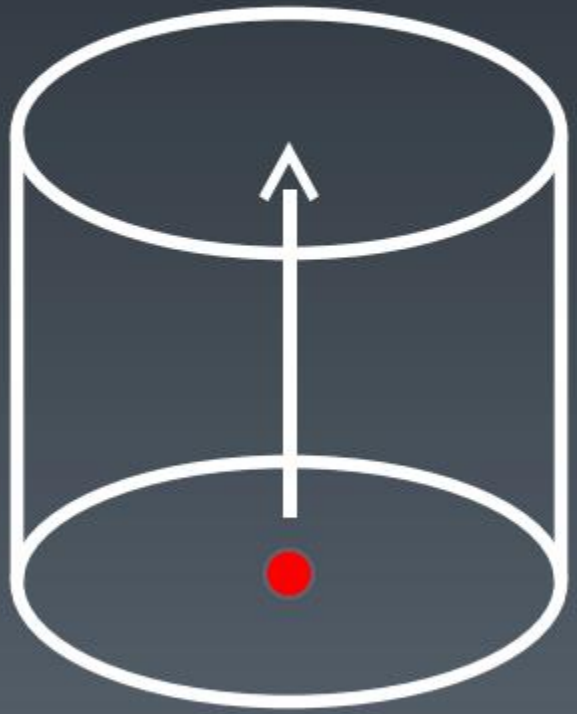
CIRCLE





Plan of the Tholos at Epidaurus

CYLINDER

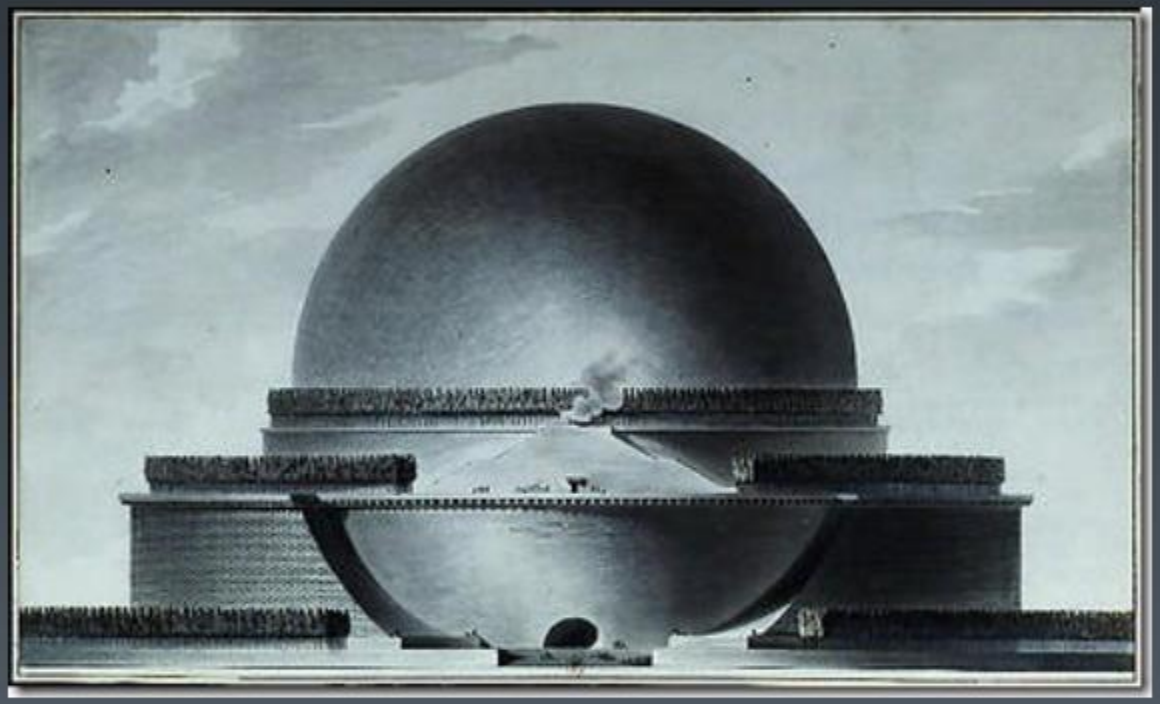




Baptistery at Pisa, Italy

SPHERE





Cenotaph for Sir Isaac Newton

LINE

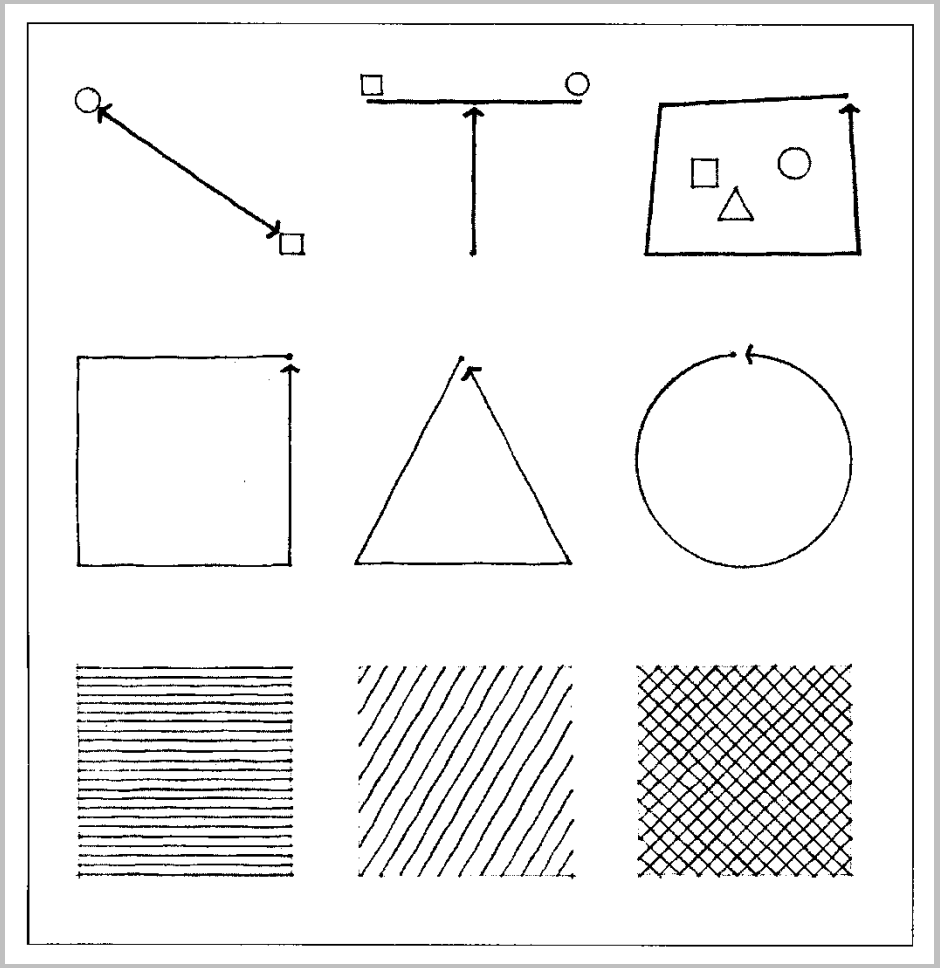
A Line is a critical element in the formation of any visual construction.

It can serve to:

- join, link, support, surround, or intersect other visual elements.

- describe the edges of and give shape to planes.

- articulate the surface of planes.



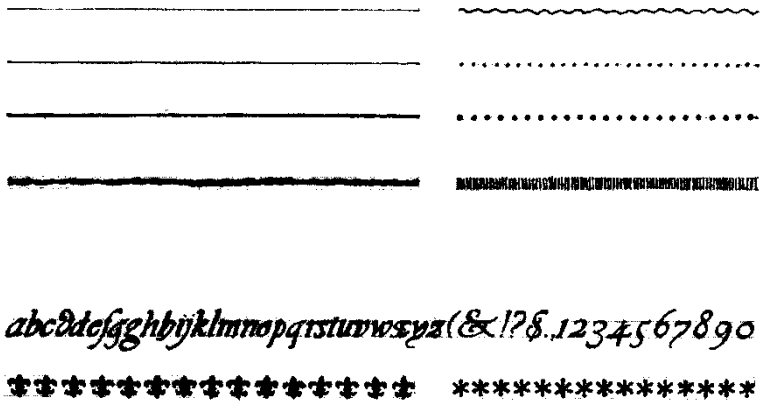
LINE

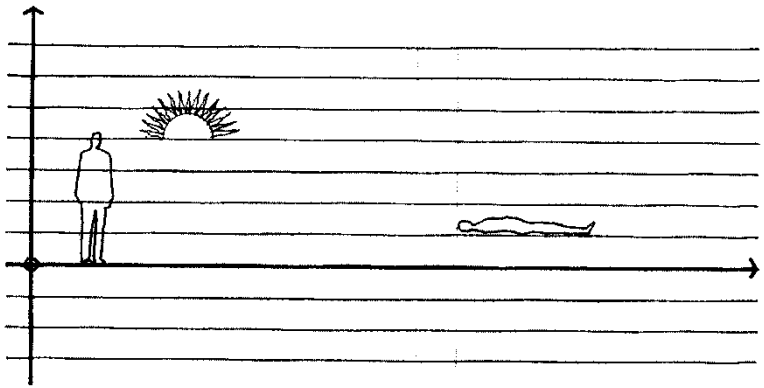
Although a line theoretically has only one dimension, it must have some degree of thickness to become visible. It is seen as a line simply because its length dominates its width. The character of a line, whether taut or limp, bold or tentative, graceful or ragged, is determined by our perception of its length- width ratio, its contour, and its degree of continuity.

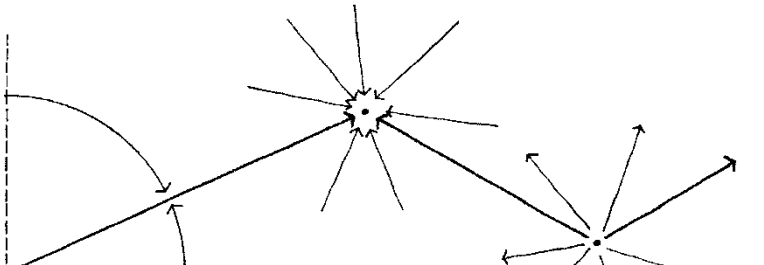
Even the simple repetition of like or similar elements, if continuous enough, can be regarded as a line. This type of line has significant textural qualities.

The orientation of a line affects its role in a visual construction. While a vertical line can express a state of equilibrium with the force of gravity, symbolize the human condition, or mark a position in space, a horizontal line can represent stability, the ground plane, the horizon, or a body at rest.

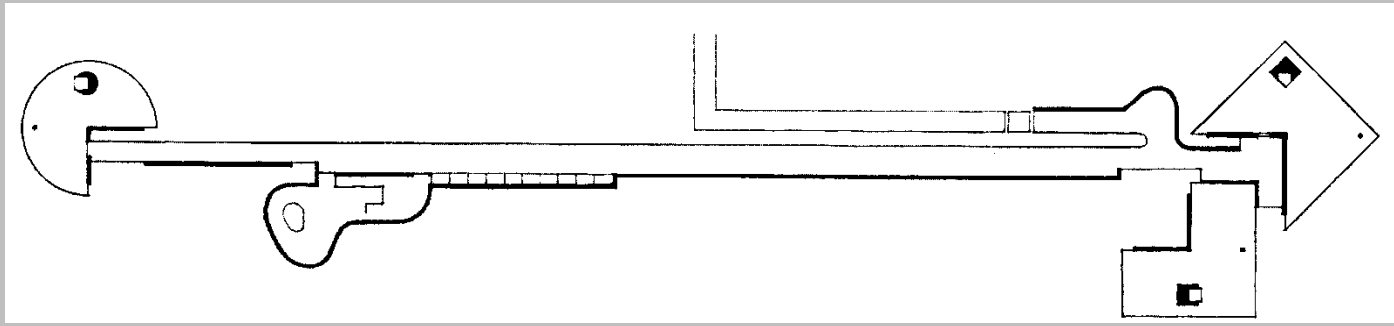
An oblique line is a deviation from the vertical or horizontal. It may be seen as a vertical line falling or a horizontal line rising. In either case, whether it is falling toward a point on the ground plane or rising to a place in the sky, it is dynamic and visually active in its unbalanced state.





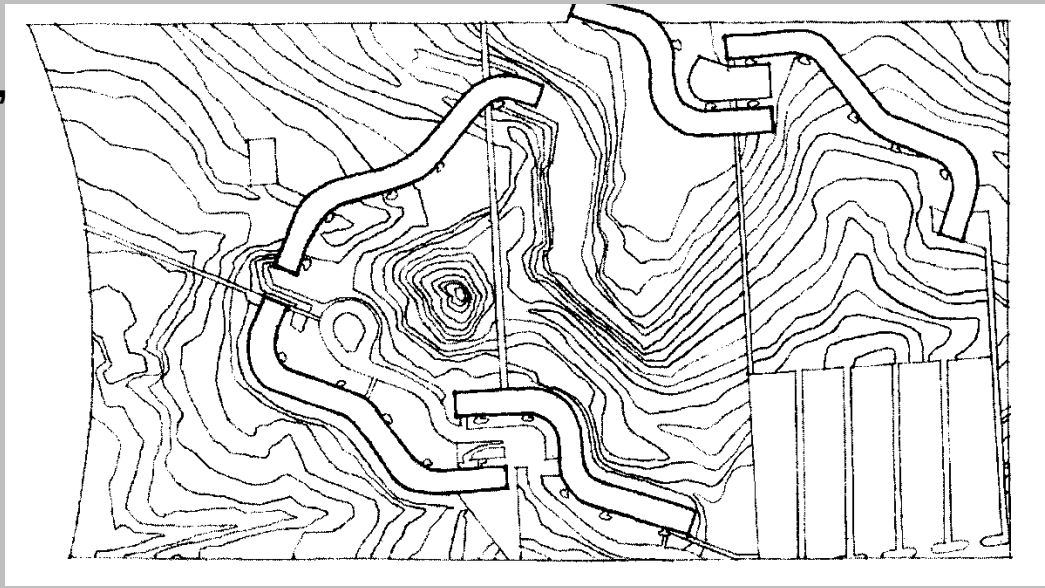


LINEAR ELEMENTS



Although architectural space exists in three dimensions, it can be linear in form to accommodate the path of movement through a building and link its spaces to one another.

Buildings also can be linear in form, particularly when they consist of repetitive spaces organized along a circulation path. As illustrated here, linear building forms have the ability enclose exterior spaces as well as adapt to the environmental conditions of a site.



LINEAR ELEMENTS



Crown Hall. School of Architecture and Urban Design, Illinois Institute of Technology, Chicago, 1956, Mies van der Rohe.

At a smaller scale, lines articulate the edges and surfaces of planes and volumes. The lines can be expressed by joints within or between building materials, by frames around window or door openings, or by a structural grid of columns and beams. How these linear elements affect the texture of a surface will depend on their visual weight, spacing, and direction .



Seagram Building. New York City, 1956-58, Mies van der Rohe and philip Johnson.

Vertical elements have been used throughout history to commemorate significant events and establish particular points in space



Column of Marcus Aurelius



Hagia Sophia, Constantinople

Vertical linear elements can also define a transparent volume of space, as in the example above, the four minarets outline a spatial field which the dome of Hagia Sophia rises in splendor

- Linear members that possess the necessary material strength can perform structural functions
- Linear elements express movement across space



Salginatobel Bridge, Switzerland

- Two parallel lines have the ability to visually describe a plane
- The closer these lines are to each other, the stronger will be the sense of plane they convey





Colonnade

PLANE

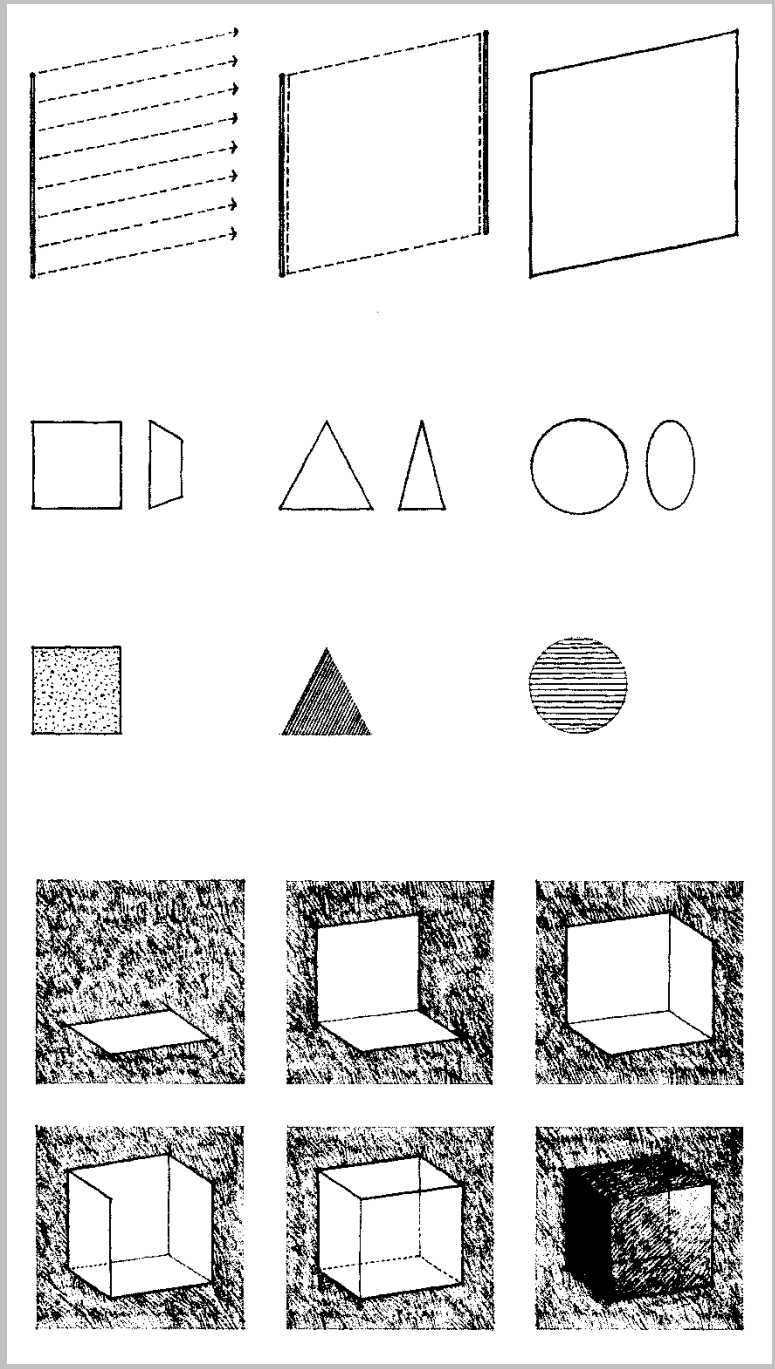
A line extended in a direction other than its intrinsic direction becomes a plane.

Conceptually, a plane has length and width, but no depth.

Shape is the primary identifying characteristic of a plane. It is determined by the contour of the line forming the edges of a plane. Because our perception of shape can be distorted by perspective foreshortening, we see the true shape of a plane only when view it frontally.

The supplementary properties of a plane—its surface color, pattern, and texture—affect its visual weight and stability.

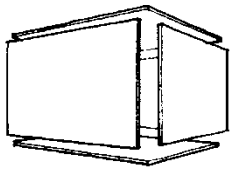
In the composition of a visual construction, a plane serves to define the limits or boundaries of a volume. If architecture as a visual art deals specifically with the formation of three-dimensional volumes of mass and space, then the plane should be regarded as a key element in the vocabulary of architectural design.

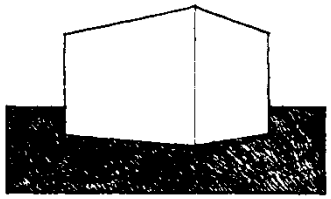


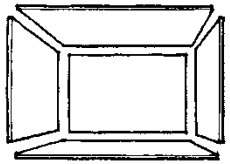
PLANE

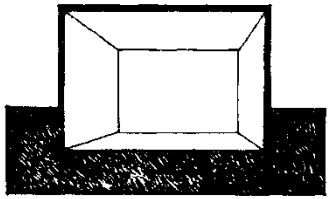
Planes in architecture define three-dimensional volumes of mass and space. The properties of each plane—size, shape, color, texture—as well as their spatial relationship to one another ultimately determine the visual attributes of the form they define and the qualities of space they enclose.

In Architectural design, we manipulate three generic types of planes:









Overhead Plane

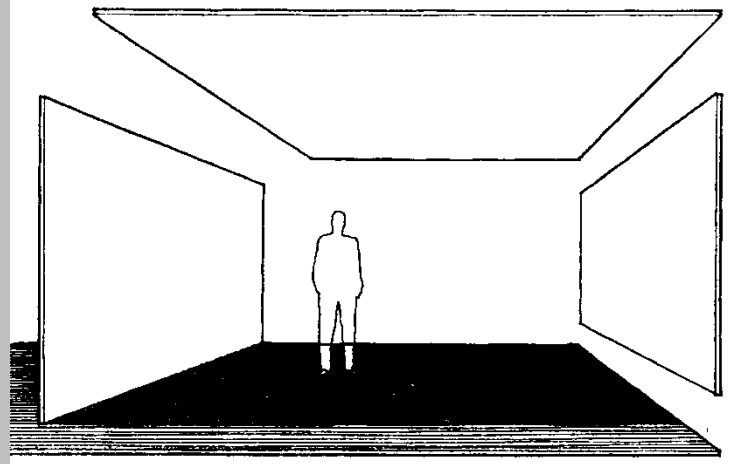
The overhead plane can be either the roof plane that shelters the interior spaces of a building from the climatic elements, or the ceiling plane that forms the upper enclosing surface of a room.

Wall Plane

The wall plane, because of its vertical orientation, is active in our normal field of vision and vital to the shaping and enclosure of architectural space.

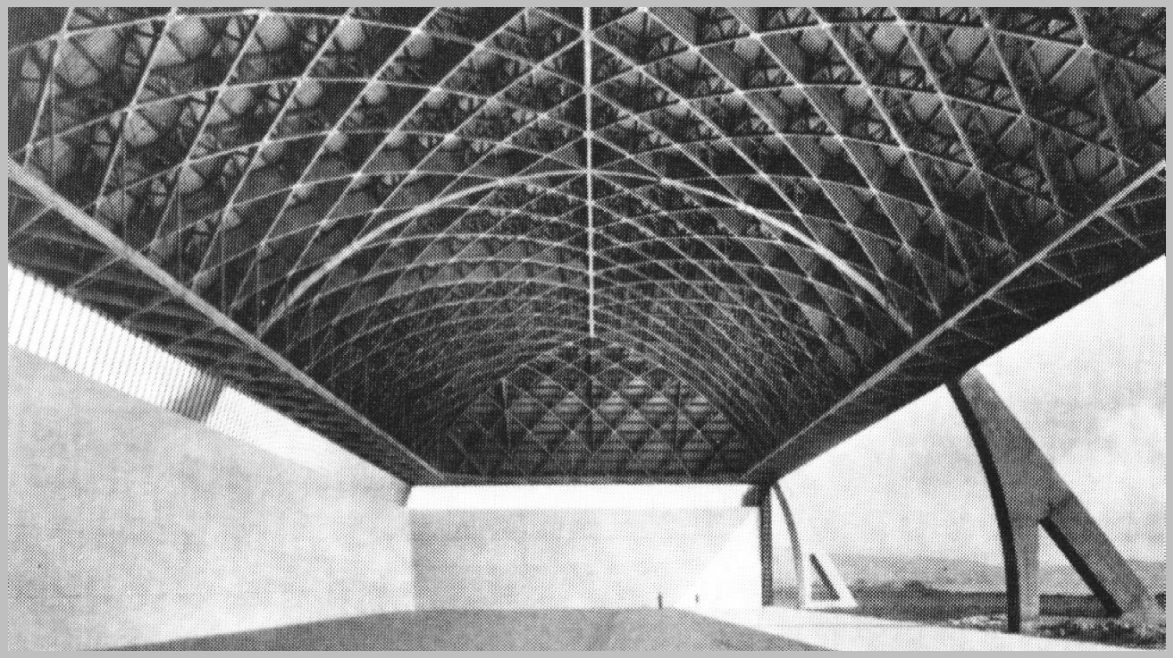
Base Plane

The base plane can be either the ground plane that serves as the physical foundation and visual base for building forms, or the floor plane that forms the lower enclosing surface of a room upon which we walk.



PLANAR ELEMENTS

While we walk on a floor and have physical contact with walls, the ceiling plane is usually out of our reach and is almost always a purely visual event in a space. It may be the underside of an overhead floor or roof plane and express the form of its structure as it spans the space between its supports, or it may be suspended as the upper enclosing surface of a room or hall.



Hangar, Design I, 1935, Pier Luigi Nervi.

The lamella structure expresses the way forces are resolved and channeled down to the roof supports.

OVERHEAD PLANE

The overhead plane can be either the roof plane that shelters the interior spaces of a building from the climatic elements, or the ceiling that forms the upper enclosing surface of the room.





WALL PLANE

The wall plane, because of its vertical orientation, is active in our normal field of vision and vital to the shaping and enclosure of architectural space.





BASE PLANE

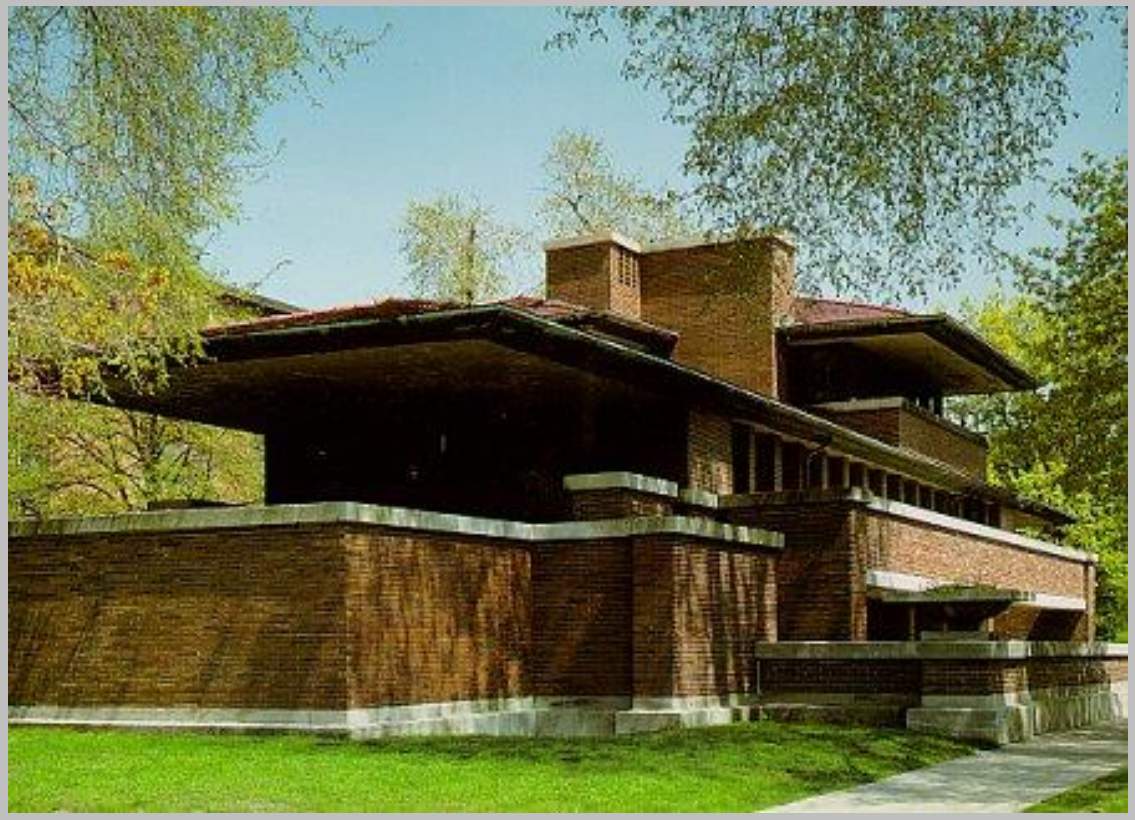
The base plane can either be ground plane that serves as the physical foundation and visual base for building forms, or the floor plane that forms the lower enclosing surface of a room upon which we walk.





PLANAR ELEMENTS

A roof plane can extend outward to form overhangs that shield door and widow openings from sun or rain, or continue downward further still to relate itself more closely to the ground plane. In warm climates, it can be elevated to allow cooling breezes to flow across and through the interior spaces of building.



Robie House, Chicago, 1909, Frank Lloyd Wright.

The low sloping roof planes and broad overhangs are characteristic of the Prairie School of Architecture.

VOLUME

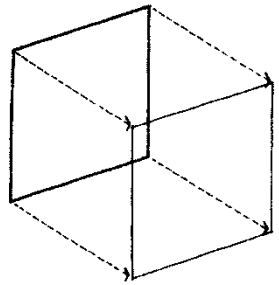
A plane extended in a direction other than its intrinsic direction becomes a volume. Conceptually, a volume has three dimensions: length, width and depth.

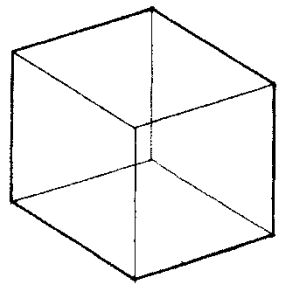
All volumes can be analyzed and understood to consist of:

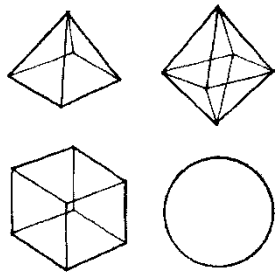
- points or vertices where several planes come together

- lines or edges where two planes meet

- planes or surfaces which define the limits or boundaries of a volume.

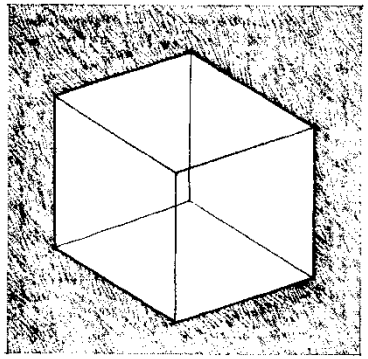


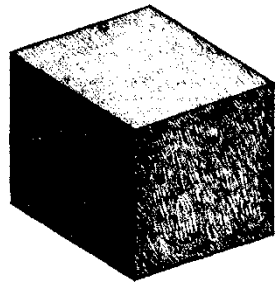




Form is the primary identifying characteristic of a volume. It established by the shapes and interrelationships of the planes that describe the boundaries of the volume.

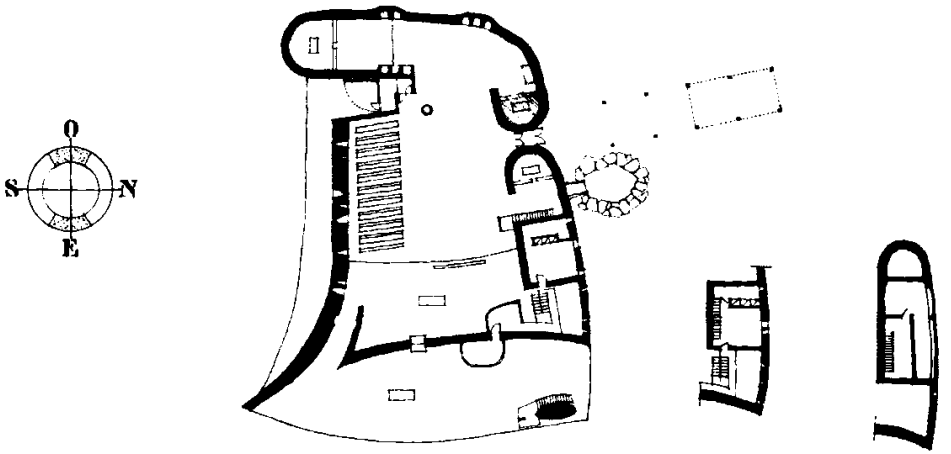
As the three-dimensional element in the vocabulary of architectural design, a volume can be either a solid—space displaced by mass—or a void—space contained or enclosed by planes.





VOLUME

In architecture, a volume can be seen to be either a portion of space contained and defined by wall, floor, and ceiling or roof planes, or a quantity of space displaced by the mass of building. It is important to perceive this duality, especially when reading orthographic plans, elevations and sections.



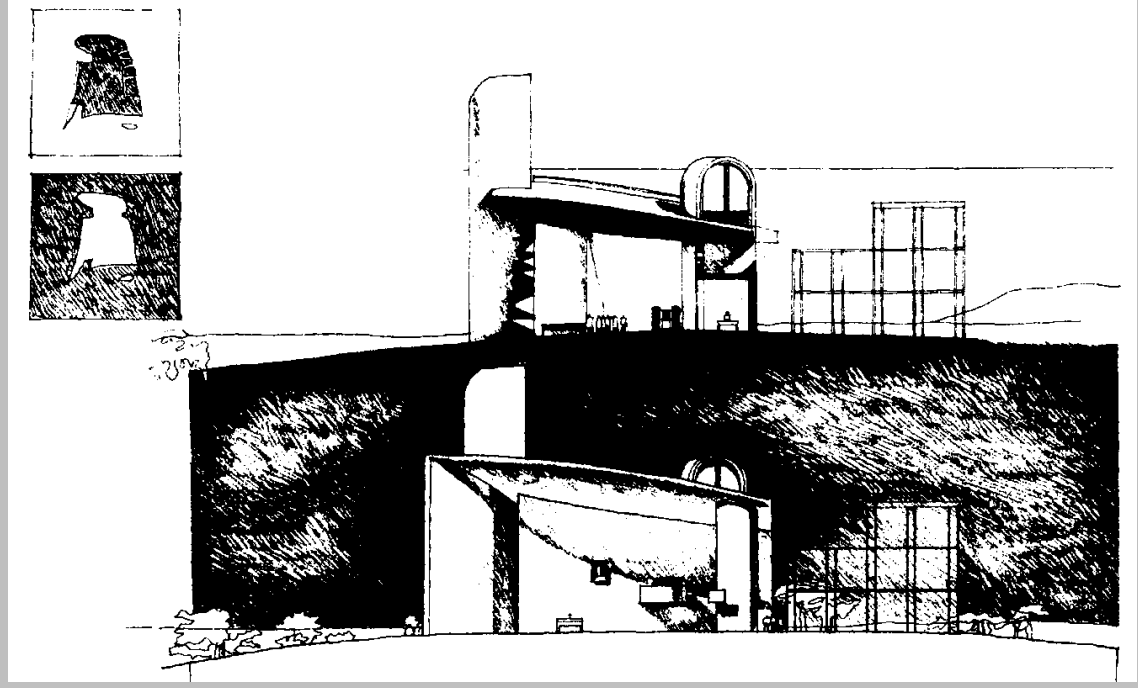
Plan and Section

Space defined by wall, floor, and ceiling or roof planes.



Elevation

Space displaced by the mass of a building.



Notre Dame Du Haut, Ronchamp, France, 1950-55, Le Corbusier.



FLOOR PLAN

space defined by wall, floor and ceiling



ELEVATION

space displaced by the mass of a building

VOLUMETRIC ELEMENTS



Villa Savoye, Poissy, France, 1928-31, Le Corbusier.

Building forms that stand as an object in the landscape can be read as occupying volume in space.

VOLUMETRIC ELEMENTS



Piazza del Campidoglio, Rome, C. 1544, Michelangelo.

A series of buildings enclose an urban square.

Building forms that serve as containers can be read as masses that define volumes of space.